Cambridge (CIE) A Level Computer Science

Your notes

Database Management Systems (DBMS)

Contents

- DBMS Features

Your notes

DBMS vs file-based systems

What is a DBMS?

- A DBMS (Database Management System) is **software** that is used to **create, manage, and interact with databases**
- It addresses the limitations of file-based systems by:

Area	Limitation in file-based systems	How a DBMS solves it
Data redundancy	Same data is stored in multiple files, wasting space and causing confusion	A DBMS **centralises data**, so it is stored once and referenced when needed
Data inconsistency	Data may differ between files if one is updated but others are not	A DBMS ensures **data consistency** by updating a single source of truth
Data management	Each application stores and manages its own data	A DBMS provides a **data dictionary** that defines all data elements across the system (field/table names, data types)
Data modelling	No structured way to represent or link data	A DBMS uses **data modelling tools** (e.g. ER diagrams) to define entities and relationships
Logical schema	No separation of how data is stored vs how it is used	A DBMS uses a **logical schema** to describe the structure independently of storage
Data integrity	Difficult to enforce validation rules across different files	A DBMS uses **constraints** (e.g. primary keys, foreign keys) to enforce integrity
Data security	Poor control over who can access or change data	A DBMS supports **user accounts, access rights, and permissions** for individuals/groups
Backup procedures	Often manual and inconsistent	A DBMS includes **automated backup and recovery tools** to reduce risk of data loss

© 2025 Save My Exams, Ltd.

DBMS tools

Your notes

Developer interface

- The developer interface allows developers to **write database queries** using **Structured Query Language (SQL)**
- This is more flexible than using **query-by-example** tools, which are visual and simplified
- Using SQL allows the developer to create **more complex and precise queries** to search, update, or manage data

Query processor

- The query processor is responsible for **handling and executing SQL queries**
- It has several key components:

Component	Function
DDL interpreter	Interprets **Data Definition Language (DDL)** commands like CREATE, ALTER, and DROP and updates the **data dictionary**
DML compiler	Compiles **Data Manipulation Language (DML)** statements (e.g. SELECT, INSERT, UPDATE) into low-level instructions and **optimises** the query
Query evaluation engine	Executes the compiled instructions to **retrieve or manipulate** the actual data in the database

- Together, these components make sure that any query written by the developer is **interpreted, optimised**, and **executed efficiently**

Worked Example

State what is meant by a data dictionary and give one example of an item typically found in a data dictionary. **[2]**

Answer

- A data dictionary is data about the data structure in the database [1 mark]
- Example: field name, data type, or table name [1 mark]